

Qualification Pack



Mining Rescuer

Opencast/ Underground Metal/ Underground Coal/ Rare Earth Plant

QP Code: MIN/Q1701

Version: 1.0

NSQF Level: 3.0

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MIN/Q1701: Mining Rescuer

Brief Job Description

This job role refers to any person who rescues people who have come under accident by fire, gaseous hazards, work at height, inundation, confined space hazards, etc.

Personal Attributes

Technical knowledge of techniques and tools required for handling hazards like fire, inundation, work at height, confined spaces. The person should have ability to plan and prioritize, quality consciousness, sensitivity to problem solving, quick decision making, safety orientation, reading, writing and communication skills, dexterity and high precision, ability to use internal ERP systems

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N1706: Undertake actions to prevent the fire situation in a mine](#)
2. [MIN/N1707: Conduct fire rescue operation](#)
3. [MIN/N1708: Conduct rescue operation from surface water inundation, underground inundation, gaseous hazards, roof fall, work at height and confined places](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Electives(mandatory to select at least one):

Elective 1: Opencast

1. [MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines](#)

Elective 2: Underground Metal

1. [MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines](#)

Elective 3: Underground Coal

1. [MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines](#)

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Elective 4: Rare Earth Plant

1. [MIN/N1705: Follow Health, Safety and Environmental guidelines for Rare Earth \(RE\) Chemical plant](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions
Country	India
NSQF Level	3.0
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8182.0800
Minimum Educational Qualification & Experience	10th grade pass OR 9th grade pass (with 1 year relevant experience) OR 8th grade pass (with 2 years relevant experience) OR 5th grade pass (with 5 years relevant experience) OR Previous relevant Qualification of NSQF Level (2.5-Mining-Ventilation Crew Personnel with 1.5 Year of experience in relevant field)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	28/09/2026
NSQC Approval Date	29/09/2023
Version	1.0
Reference code on NQR	QG-03-MG-01027-2023-V2-SCMS

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NQR Version	1.0
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Remarks:

Total Duration: 420 Hours (360 hours Domain + 60 Hours Employability Skills)

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MIN/N1706: Undertake actions to prevent the fire situation in a mine

Description

This OS unit is about identifying the organizational requirements to ensure safety from fire and undertaking the actions to prevent the fire situations in a mine

Scope

The scope covers the following :

- Identify the organizational requirements to ensure safety from fire
- Undertake actions to prevent the fire situations in a mine

Elements and Performance Criteria

Identify the requirements to ensure fire safety in the mines

To be competent, the user/individual on the job must be able to:

- PC1.** identify the common types of fires i.e. accidental fires and spontaneous fires as well as the possible sources of the same
- PC2.** identify the sources of fire i.e. diesel engine, oil spillage, electrical fires, frictional fires, blasting, explosion etc
- PC3.** identify the potential location/s of fire i.e. compressors, explosive stores etc
- PC4.** identify various ways to extinguish the different types of fires and associated equipment required like water sprinklers, fire extinguishers, etc , degree of gassiness and its permissible apparatus, prevent and outbreak belowground or the spread to one part to other part of mine of adjoining mine, all unused workings connected to surface though walkable entrance to be inspected, periodic checks of sealed off areas due to fire breakouts, in open cast working all herbaceous plants shall be removed and all dead leaves and dry vegetation cleared off
- PC5.** check the availability and workability of fire alarms at the site
- PC6.** check the adequacy and availability of fire extinguishing equipment at the site

Undertake actions to prevent the fire situations in a mine

To be competent, the user/individual on the job must be able to:

- PC7.** deal with shortfalls that affect the availability of resources to meet operational readiness
- PC8.** facilitate the awareness of fire safety and associated safety actions that could be taken at the time of fire in the mine
- PC9.** enter inspection, training and emergency medical service calls into the records
- PC10.** ensure adherence to the fire safety related policies and Guidelines defined for the organization
- PC11.** ensure periodic checking and maintenance of the fire extinguishing equipment at the site

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** different types of mines and detail of the mine he is working in-classification of gassy seam into degree of gassiness; Adequate and sufficient arrangements for early detection , control and extinguishing fire; Measures and precautions with regard to appropriate nature of mine operation to prevent, detect and combat of mine fires as prescribed by mine officials
- KU2.** mine organization, time keeping, need for discipline and punctuality
- KU3.** benching in quarries, dressing of overhangs, undercuts, fencing, first aid and hygiene
- KU4.** standing orders in force at the mine
- KU5.** shot-firing and safety regulations
- KU6.** how and where to take shelter, in case of threat to the safety and health of workers
- KU7.** provision of wages, working hours and accident compensation as per Mines act
- KU8.** mining safety procedures
- KU9.** impact of violation of safety procedures
- KU10.** standard operating procedure (SOP) followed by the company
- KU11.** processes like fire fighting, procurement, store management, inventory management, quality management and key contact points for query resolution
- KU12.** different types of sources of fire like frictional fires, electrical fires, explosives, oil spills, blasting, etc.
- KU13.** different types of fire safety equipment like fire extinguishers, water sprinklers, breathing apparatus, fire jackets, fire alarms etc
- KU14.** hazards and safety aspects involved in fire prevention and rescue activities and usage of relevant PPEs
- KU15.** mines rescue rules 1985
- KU16.** mine plan, goaf area, de-pillaring sites , stone dust barriers, checking the unused, isolated areas for any leftovers
- KU17.** mock drills
- KU18.** safety in the vicinity of machinery
- KU19.** bound paged book kept for inspection purpose duly signed and dated by inspection person

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any) related to the fire fighting
- GS2.** write information documents to internal departments/teams or enter the information in online ERP systems under guidance of the supervisor
- GS3.** read equipment manuals and process documents to understand the equipment and processes better
- GS4.** read internal information documents sent by internal teams
- GS5.** read and execute prescribed measures and precautions appropriate to the nature of mine by owner, agent or manager
- GS6.** discuss task lists, schedules and activities with the supervisor
- GS7.** effectively communicate with the team members



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- GS8.** discuss with supervisor in order to understand the nature of the problem and to clarify queries
- GS9.** comprehend the information given by the speaker
- GS10.** comprehend the various hand signals used
- GS11.** plan and organize the work order and jobs received from the operator
- GS12.** complete the job defined by the supervisor within timelines and quality norms
- GS13.** detect problems in day to day tasks
- GS14.** support supervisor in using specific problem solving techniques and detailing out the problems
- GS15.** discuss possible solution with the supervisor for problem solving
- GS16.** make decisions in emergency conditions in case the supervisor is not available(as per the authority matrix defined by the organization)

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify the requirements to ensure fire safety in the mines</i>	15	25	-	10
PC1. identify the common types of fires i.e. accidental fires and spontaneous fires as well as the possible sources of the same	2	4	-	2
PC2. identify the sources of fire i.e. diesel engine, oil spillage, electrical fires, frictional fires, blasting, explosion etc	2	4	-	2
PC3. identify the potential location/s of fire i.e. compressors, explosive stores etc	3	4	-	1
PC4. identify various ways to extinguish the different types of fires and associated equipment required like water sprinklers, fire extinguishers, etc , degree of gassiness and its permissible apparatus, prevent and outbreak belowground or the spread to one part to other part of mine of adjoining mine, all unused workings connected to surface though walkable entrance to be inspected, periodic checks of sealed off areas due to fire breakouts, in open cast working all herbaceous plants shall be removed and all dead leaves and dry vegetation cleared off	3	4	-	1
PC5. check the availability and workability of fire alarms at the site	2	4	-	2
PC6. check the adequacy and availability of fire extinguishing equipment at the site	3	5	-	2
<i>Undertake actions to prevent the fire situations in a mine</i>	15	25	-	10
PC7. deal with shortfalls that affect the availability of resources to meet operational readiness	3	5	-	2
PC8. facilitate the awareness of fire safety and associated safety actions that could be taken at the time of fire in the mine	3	5	-	2
PC9. enter inspection, training and emergency medical service calls into the records	3	5	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. ensure adherence to the fire safety related policies and Guidelines defined for the organization	3	5	-	2
PC11. ensure periodic checking and maintenance of the fire extinguishing equipment at the site	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1706
NOS Name	Undertake actions to prevent the fire situation in a mine
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions
NSQF Level	3.0
Credits	2
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	28/09/2026
NSQC Clearance Date	29/09/2023

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MIN/N1707: Conduct fire rescue operation

Description

This OS unit is about conducting firefighting operations which comprise of fire control as well as fire rescue

Scope

The scope covers the following :

- Conduct the fire fighting and rescue operations

Elements and Performance Criteria

Conduct Fire Rescue Operations

To be competent, the user/individual on the job must be able to:

- PC1.** equip himself with the fire protection apparatus including breathing apparatus, fire jackets, hoses etc. and ensure adequate numbers of persons are accompanying
- PC2.** cut the power supply in the mine
- PC3.** use various equipment to control fire including ladder trucks, pumper trucks, tanker trucks, fire hose, fire extinguishers etc
- PC4.** conduct fire control by depriving a fire of fuel (Reducing Agent), oxygen (Oxidizing Agent), heat and/or the chemical chain reaction that are necessary to sustain itself or re-kindle
- PC5.** conduct preliminary salvage operations such as throwing salvage covers, sweeping water and removing debris
- PC6.** attack the mine fires through hose streams channelized towards fire
- PC7.** contact fire brigade/ rescue team immediately for extended support
- PC8.** search and remove the trapped occupants out of hazardous conditions
- PC9.** conduct search that is primarily around movement against room walls (to prevent rescuers from becoming lost or disoriented) and methodical searches of specific areas by designated teams
- PC10.** check and mobilize the resources from adjoining areas
- PC11.** inspect and clean the equipment and apparatus after returning from a fire
- PC12.** make minor repairs to equipment, perform routine preventive maintenance tasks, and keep records of such action when unable to locate fire report to senior officials and carry out intentional flooding with their authorization

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of mines and detail of the mine he is working in- classification of gassy seam into degree of gassiness; adequate and sufficient arrangements for early detection , control and extinguishing fire; Measures and precautions with regard to appropriate nature of mine operation to prevent detect and combat of mine fires as prescribed by mine officials

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- KU2.** mine organization, time keeping, need for discipline and punctuality
- KU3.** benching in quarries, Dressing of overhangs, Undercuts, Fencing, First aid and Hygiene
- KU4.** standing orders in force at the mine
- KU5.** shot-firing and safety regulations
- KU6.** evacuation to safe location threatening the safety and health of workers
- KU7.** provision of wages, working hours and accident compensation as per Mines act
- KU8.** mining safety procedures
- KU9.** impact of violation of safety procedures
- KU10.** standard operating procedures (SOP) followed in the company
- KU11.** processes like firefighting, procurement, store management, inventory management, quality management and key contact points for query resolution
- KU12.** different types of sources of fire like frictional fires, electrical fires, explosives, oil spills, blasting, etc.
- KU13.** different types of fire safety equipment like fire extinguishers, water sprinklers, breathing apparatus, fire jackets, fire alarms etc. and their usage
- KU14.** hazards and safety aspects involved in fire prevention and rescue activities and usage of relevant PPEs
- KU15.** mines rescue rules 1985
- KU16.** mine plan, goaf area, de-pillaring sites sealed of areas, etc.
- KU17.** mock drills
- KU18.** for adequate precautions to prevent danger from noxious, asphyxiating or inflammable gasses
- KU19.** danger faced at site (protection from flame, steam ejected or rolling down of hot materials)
- KU20.** how to ensure safety of persons engaged in various operations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** document information from the sketches and engineering drawings
- GS2.** prepare draft drawings for the final output product
- GS3.** note down observations (if any) related to the fire fighting process
- GS4.** write information documents to internal departments/ internal teams or enter the information in online ERP systems under Guidance of the supervisor
- GS5.** enter the information in online ERP systems under Guidance of the supervisor
- GS6.** read and interpret engineering drawing and sketches
- GS7.** read and interpret symbols and measurements used in the drawings
- GS8.** read equipment manuals and process documents to understand the equipment and processes better
- GS9.** read internal information documents sent by internal teams
- GS10.** discuss task lists, schedules and activities with the supervisor
- GS11.** effectively communicate with the team members

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- GS12.** discuss with the supervisor in order to understand the nature of the problem and to clarify queries
- GS13.** comprehend the information given by the speaker
- GS14.** interpret the various hand signals used
- GS15.** plan and organize the work order and jobs received from the operator
- GS16.** organize all process/ equipment manuals so that sorting/ accessing information is easy
- GS17.** ensuring safety of persons engaged in said operations
- GS18.** take measures and precautions in regard to appropriate nature of mine operation to prevent, detect and combat
- GS19.** use common sense and make judgments during day to day basis
- GS20.** use reasoning skills to identify and resolve basic problems
- GS21.** follow instructions and work on areas of improvement identified complete the assigned tasks with minimum supervision
- GS22.** complete the job defined by the supervisor within the timelines and quality norms
- GS23.** encouraging persons employed in mines not to place, throw any naked light, timber, wooden structures or other combustible materials other than designated areas

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct Fire Rescue Operations</i>	30	50	-	20
PC1. equip himself with the fire protection apparatus including breathing apparatus, fire jackets, hoses etc. and ensure adequate numbers of persons are accompanying	2	4	-	1
PC2. cut the power supply in the mine	2	4	-	1
PC3. use various equipment to control fire including ladder trucks, pumper trucks, tanker trucks, fire hose, fire extinguishers etc	2	5	-	2
PC4. conduct fire control by depriving a fire of fuel (Reducing Agent), oxygen (Oxidizing Agent), heat and/or the chemical chain reaction that are necessary to sustain itself or re-kindle	2	5	-	2
PC5. conduct preliminary salvage operations such as throwing salvage covers, sweeping water and removing debris	2	4	-	1
PC6. attack the mine fires through hose streams channelized towards fire	2	4	-	1
PC7. contact fire brigade/ rescue team immediately for extended support	3	4	-	2
PC8. search and remove the trapped occupants out of hazardous conditions	3	4	-	2
PC9. conduct search that is primarily around movement against room walls (to prevent rescuers from becoming lost or disoriented) and methodical searches of specific areas by designated teams	3	4	-	2
PC10. check and mobilize the resources from adjoining areas	3	4	-	2
PC11. inspect and clean the equipment and apparatus after returning from a fire	3	4	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. make minor repairs to equipment, perform routine preventive maintenance tasks, and keep records of such action when unable to locate fire report to senior officials and carry out intentional flooding with their authorization	3	4	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1707
NOS Name	Conduct fire rescue operation
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions
NSQF Level	3.0
Credits	4
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	28/09/2026
NSQC Clearance Date	29/09/2023

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MIN/N1708: Conduct rescue operation from surface water inundation, underground inundation, gaseous hazards, roof fall, work at height and confined places

Description

This OS unit is about conducting rescue operation from surface water inundation, underground inundation, gaseous hazards, roof fall, work at height and confined places

Scope

The scope covers the following :

- Instruct the rescue procedures
- Rescue from surface water and underground inundation
- Rescue from gaseous hazards
- Rescue against roof fall, work at height and confined places

Elements and Performance Criteria

Instruct the rescue procedures

To be competent, the user/individual on the job must be able to:

- PC1.** establish a fresh air base from where the rescue operation to be conducted
- PC2.** proceed with self contained breathing apparatus and appropriate rescue equipment
- PC3.** work as team player or leader of rescue team, as per requirement
- PC4.** return to the fresh air base (with whole team), if any member experiences trouble due to physical exhaustion or defect in apparatus
- PC5.** communicate with hand signals and coded sounds of horns
- PC6.** mark the route followed by the team (smoky and poor visibility)
- PC7.** use life line extended from the base as the team proceeds

Rescue from surface water and underground inundation

To be competent, the user/individual on the job must be able to:

- PC8.** identify the source of inundation and inform the workers to withdraw themselves
- PC9.** identify an escape route which is free from depression, deep drains, hand rails
- PC10.** identify the places of underground inundation (seam or section below, water bearing strata)
- PC11.** wear scuba set, and use it to rescue equipment, personnel, and other purposes.
- PC12.** dewater the water logged workings in underground/opencast and give immediate strong support to roof and sides

Rescue from gaseous hazards

To be competent, the user/individual on the job must be able to:

- PC13.** ensure proper ventilation upto the face and conduct frequent checks for methane, oxygen, carbon monoxide, hydrogen sulphides, etc.
- PC14.** use approved flame safety lamp

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- PC15.** initiate withdrawal of people by shortest safe route whether ordinarily used as travelling roadway or not
 - PC16.** clear underground roads blocked by roof falls, tubs, mine cars, machinery and materials , twisted rails cables etc
 - PC17.** remove gas through roof cavity
- Rescue against roof fall, work at height and confined places*
- To be competent, the user/individual on the job must be able to:
- PC18.** assess the physical condition of the debris of roof fallen and the people and equipment trapped, if any
 - PC19.** ensure power is cut and fencing of the site is done
 - PC20.** use proper rescue equipment to remove the debris
 - PC21.** use anchor points, slings, ropes, full body harness, extra pulley, and other equipment, for rescuing person trapped while working at height.
 - PC22.** use ropes, communication system, and other equipment for rescuing person trapped in confined space

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** capacity of the rescue apparatus , fresh air base and its time to time advancing as exploration proceeds,
- KU2.** experienced staff and workers to be deployed, capacity of the team to work and cover the distance and load
- KU3.** the nature of duties expected to perform by the team, temperature of the underground atmosphere the gradient of roadway underground/opencast workings
- KU4.** roles and responsibilities of rescue trained personnel
- KU5.** different types of mine working (each section/districts, working at height), common geological disturbances like Dykes, fault, sill etc
- KU6.** provisions regarding depillaring and different method (stook, slicing), mine gases, their detection, permissible limits, eruption, mode of emission of firedamps and also dangers of firedamp explosions
- KU7.** safety equipments and application of a sinking shaft, dangers from coal dust, permissible limits of respirable dust, defense mechanism of human body and precaution
- KU8.** all useful information including position, extent and depth of features, shall be acquired, development within of water logged working, dewatering, permission granted by regional inspector and the scheme of working, the record showing exact height and width of number of boreholes driven, diameter and length of each bore hole, places at which and the direction in which each bore hole was driven
- KU9.** surface sources of inundation (flowing waters-rivers canals) (standing water-ponds, tanks) (accumulated water-in low lying area, subsidence area and abandoned in underground (UG) as well as opencast (OC) working), preventing inundation from surface sources, constant monitoring of highest flood level (HFL), embankment , water level closed or abandoned shafts, subsidence etc.

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- KU10.** assessment of danger, action plan, standing order against inundation (giving warning, unusual seepage, flood warnings) water danger plan, fire stoppings, isolation stoppings
- KU11.** process to remove gas from Roof cavity, bleeding gas from isolated old working, layering of methane, limits of inflammability, against Fire damp explosion, Backlash, Rescue operation at the mine, Rescue station etc
- KU12.** self contained breathing apparatus and its operation, usages of rescue apparatus like self rescue kit, gas mask, PPE, reviving apparatus, etc.
- KU13.** different method of Resuscitation and its usages (Schafer's, Sylvester method, Mouth to Mouth resuscitation),
- KU14.** the danger and difficulties in rescuing survivors after an explosion (shaft, cages may be damaged or unsafe to use, smoke and dust reducing visibility, fumes, noxious gases and after damp are fatal to the men attempting to rescue the survivors, Telephonic lines disruption, surface fan may be damaged disrupting the mine ventilation)
- KU15.** flame safety lamp
- KU16.** usage anchor points, slings, ropes, full body harness, extra pulley, and other equipment

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain daily maintenance checklist and other documents related to troubleshooting
- GS2.** check and record in different recording systems such as bound book, ERP systems
- GS3.** adhere to safety management plan, SOP finalized by the regulatory authority
- GS4.** report for any discrepancy to senior authority
- GS5.** report unidentified problems to the competent senior persons
- GS6.** keep record of useful information of position, extent and depth of structural, geological features

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Instruct the rescue procedures</i>	10	17	-	6
PC1. establish a fresh air base from where the rescue operation to be conducted	2	3	-	1
PC2. proceed with self contained breathing apparatus and appropriate rescue equipment	2	3	-	1
PC3. work as team player or leader of rescue team, as per requirement	1	2	-	1
PC4. return to the fresh air base (with whole team), if any member experiences trouble due to physical exhaustion or defect in apparatus	1	2	-	-
PC5. communicate with hand signals and coded sounds of horns	1	2	-	1
PC6. mark the route followed by the team (smoky and poor visibility)	2	3	-	1
PC7. use life line extended from the base as the team proceeds	1	2	-	1
<i>Rescue from surface water and underground inundation</i>	9	13	-	4
PC8. identify the source of inundation and inform the workers to withdraw themselves	1	2	-	1
PC9. identify an escape route which is free from depression, deep drains, hand rails	2	2	-	1
PC10. identify the places of underground inundation (seam or section below, water bearing strata)	2	3	-	1
PC11. wear scuba set, and use it to rescue equipment, personnel, and other purposes.	2	3	-	-
PC12. dewater the water logged workings in underground/opencast and give immediate strong support to roof and sides	2	3	-	1
<i>Rescue from gaseous hazards</i>	6	10	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. ensure proper ventilation upto the face and conduct frequent checks for methane, oxygen, carbon monoxide, hydrogen sulphides, etc.	1	2	-	1
PC14. use approved flame safety lamp	1	2	-	1
PC15. initiate withdrawal of people by shortest safe route whether ordinarily used as travelling roadway or not	2	2	-	1
PC16. clear underground roads blocked by roof falls, tubs, mine cars, machinery and materials , twisted rails cables etc	1	2	-	1
PC17. remove gas through roof cavity	1	2	-	1
<i>Rescue against roof fall, work at height and confined places</i>	5	10	-	5
PC18. assess the physical condition of the debris of roof fallen and the people and equipment trapped, if any	1	2	-	1
PC19. ensure power is cut and fencing of the site is done	1	2	-	1
PC20. use proper rescue equipment to remove the debris	1	2	-	1
PC21. use anchor points, slings, ropes, full body harness, extra pulley, and other equipment, for rescuing person trapped while working at height.	1	2	-	1
PC22. use ropes, communication system, and other equipment for rescuing person trapped in confined space	1	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1708
NOS Name	Conduct rescue operation from surface water inundation, underground inundation, gaseous hazards, roof fall, work at height and confined places
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions
NSQF Level	3.0
Credits	4
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	28/09/2026
NSQF Clearance Date	29/09/2023

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines

Description

This unit is about adhering to health, safety and environmental guidelines at the Open Cast Mines while executing various tasks, maintenance and operations.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow Environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** comply with safety, health, and security-related regulations/guidelines at the opencast mine
- PC2.** follow the safety instructions given by the workman's inspector
- PC3.** follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.
- PC4.** take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations
- PC5.** follow appropriate Safe Operating Procedure (SOP) while dealing with explosives
- PC6.** respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility
- PC7.** provide first aid to an injured person
- PC8.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC9.** use appropriate PPE as per the requirement
- PC10.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC11.** disinfect the machine/tools before and after work/task
- PC12.** maintain hygiene at the work site
- PC13.** report any symptoms of illness to the shift-incharge

Follow Environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC14.** identify the environmental impact of related opencast mining operations
- PC15.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC16.** ensure not to mix topsoil with waste in day to day tasks
- PC17.** ensure that HEMM is washed at the designated location

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- PC18.** ensure the productivity of the machine for material/fuel conservation
- PC19.** follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)
- PC20.** assist supervisor for reducing environmental impact caused due to related mining operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** about various environmental awareness program related to mining, organized by the various government bodies/company
- KU2.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU3.** how to identify six-directional hazards
- KU4.** basic mining terminologies and definitions
- KU5.** importance of blind spot survey/training
- KU6.** means of access and egress from the mines, location of workshop, haul roads and working face including dump yards
- KU7.** duties of workers under The Mines act-1952
- KU8.** working hours and accident compensation as per The Mines act-1952
- KU9.** about precautions to be taken when handling heavy equipment
- KU10.** various problems/incidents likely to occur
- KU11.** hierarchy of the reporting
- KU12.** machine operation, condition of the machine and worksite
- KU13.** proper documents specific to the machine
- KU14.** role of workmen inspector, safety committee and internal safety organization
- KU15.** the process of top soil removal and management
- KU16.** mine sump and pumping system of the mines
- KU17.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU18.** common sources of pollution in the mines and ways to minimize it
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, gas detector, safety lamp, self-contained breathing apparatus, gum boots, ear plugs, face mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization
- KU22.** signages, mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** basic personal and workplace hygiene
- KU25.** importance of sensitization towards different genders and PWD (Persons with Disabilities)

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS4.** plan and organize the work order and jobs
- GS5.** use reasoning skills to identify and resolve fundamental problems
- GS6.** complete the assigned tasks timely

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	20	34	-	15
PC1. comply with safety, health, and security-related regulations/guidelines at the opencast mine	2	2	-	1
PC2. follow the safety instructions given by the workman's inspector	1	3	-	1
PC3. follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.	3	3	-	1
PC4. take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations	1	3	-	1
PC5. follow appropriate Safe Operating Procedure (SOP) while dealing with explosives	2	3	-	2
PC6. respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility	2	3	-	1
PC7. provide first aid to an injured person	2	3	-	1
PC8. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	3	-	1
PC9. use appropriate PPE as per the requirement	2	4	-	2
PC10. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	2	-	1
PC11. disinfect the machine/tools before and after work/task	1	2	-	1
PC12. maintain hygiene at the work site	1	1	-	1
PC13. report any symptoms of illness to the shift-incharge	1	2	-	1
<i>Follow Environmental guidelines</i>	10	16	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify the environmental impact of related opencast mining operations	2	2	-	1
PC15. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	2	-	1
PC16. ensure not to mix topsoil with waste in day to day tasks	2	2	-	1
PC17. ensure that HEMM is washed at the designated location	2	2	-	1
PC18. ensure the productivity of the machine for material/fuel conservation	1	3	-	-
PC19. follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)	1	3	-	1
PC20. assist supervisor for reducing environmental impact caused due to related mining operations	1	2	-	-
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1703
NOS Name	Follow Health, Safety, and Environmental Guidelines for opencast mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, , Exploration, Mineral Estimation, Planning, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Opencast, Specialized Operations, Electrical Services, Mechanical Services, Instrumentation and Control Systems, Ore Processing
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQF Clearance Date	11/03/2026

Qualification Pack

MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Metalliferous Mines (UMM) during the execution of various tasks and operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC2.** comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine
- PC3.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC4.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC5.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC6.** use self-rescue apparatus, appropriately when required
- PC7.** follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.
- PC8.** follow precautions against U/G electrical appliances
- PC9.** follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine
- PC10.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC11.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC12.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC13.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC14.** follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety
- PC15.** follow process for reporting any unsafe act/condition in work area to the concerned person

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- PC16.** use underground mine communication system
- PC17.** ensure positive isolation near the work place if applicable
- PC18.** use appropriate PPE as per the requirement
- PC19.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC20.** disinfect the machine/tools before and after work/task
- PC21.** maintain hygiene at the work site
- PC22.** report any symptoms of illness to the shift-incharge
- PC23.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC24.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC25.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC26.** ensure that the stowing practices produce minimum disturbance to the surface
- PC27.** ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place
- PC28.** ensure the productivity of the machine for material/fuel conservation
- PC29.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC30.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases found in the mine and their effect
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (IME & PME) of person employed as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** importance of blind spot survey/training
- KU11.** about different types of machinery used in U/G mines
- KU12.** different types of supporting systems used in U/G mines as per SSP and SSR
- KU13.** about precautions to be taken when handling heavy equipment
- KU14.** various problems/incidents likely to occur
- KU15.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS

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- KU16.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU17.** common sources of pollution in the mines and ways to minimize it
- KU18.** various types of fire extinguishers
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization as per DGMS guideline
- KU22.** mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU25.** rules and regulations for safety and security while handling hazardous materials
- KU26.** safety appliances and rescue equipment
- KU27.** importance and use of various communication system used in UG mines
- KU28.** importance of positive isolation at working site
- KU29.** safety and occupational health policy of organisation
- KU30.** six directional hazard identification process
- KU31.** basic personal and workplace hygiene
- KU32.** importance of FAB (Fresh Air Base)
- KU33.** basic provisions in Mines Creche Rules, 1966 (MCR) for females employed in the mines
- KU34.** the role and responsibilities of rescue room and rescue station and how to contact them in case of emergency
- KU35.** importance of taking shelter at the miner's station during blasting operation
- KU36.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU37.** importance of following infection control policies, '5-S' practices, and waste management
- KU38.** importance of water/material/energy conservation and management
- KU39.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from machine
- GS6.** use basic applications of a computer
- GS7.** plan and organize the work order and tasks
- GS8.** use reasoning skills to identify and resolve fundamental problems

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GS9. complete the assigned tasks timely

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	42	-	17
PC1. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	-	2	-	2
PC2. comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine	1	3	-	-
PC3. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	4	-	1
PC4. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	1	1	-	-
PC5. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	1	2	-	1
PC6. use self-rescue apparatus, appropriately when required	-	4	-	1
PC7. follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.	1	2	-	1
PC8. follow precautions against U/G electrical appliances	2	2	-	2
PC9. follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine	2	2	-	1
PC10. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	2	4	-	-
PC11. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	1	1	-	-
PC13. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	-	-	-
PC14. follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety	1	2	-	1
PC15. follow process for reporting any unsafe act/condition in work area to the concerned person	-	2	-	1
PC16. use underground mine communication system	-	1	-	-
PC17. ensure positive isolation near the work place if applicable	1	1	-	1
PC18. use appropriate PPE as per the requirement	2	4	-	2
PC19. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	1	-	1
PC20. disinfect the machine/tools before and after work/task	1	1	-	-
PC21. maintain hygiene at the work site	1	1	-	1
PC22. report any symptoms of illness to the shift-incharge	1	-	-	-
PC23. identify six directional hazards at workplace and take decisions accordingly	2	-	-	-
<i>Follow environmental guidelines</i>	6	8	-	3
PC24. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	2	-	-
PC25. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	-	1	-	-
PC26. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place	1	-	-	1
PC28. ensure the productivity of the machine for material/fuel conservation	1	1	-	-
PC29. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	1	-	1
PC30. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	2	-	1
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1702
NOS Name	Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQF Clearance Date	25/02/2026

Qualification Pack

MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Coal Mines (UCM) during execution of various tasks, operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** follow preventive measures against firedamp, whitedamp, blackdamp etc.
- PC2.** use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)
- PC3.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC4.** comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine
- PC5.** ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle
- PC6.** ensure that every instrument, apparatus and equipment are DGMS approved before these are used
- PC7.** ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining
- PC8.** provide first aid to an injured person
- PC9.** follow safety precautions against spontaneous heating of the coal
- PC10.** operate various types of fire extinguishers to control different types of fire at worksite, if required
- PC11.** ensure that no person is traveling/working/staying under unsupported roof
- PC12.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC13.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC14.** use self-rescue apparatus appropriately when required
- PC15.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC16.** follow precautions against U/G electrical appliances

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- PC17.** take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports
- PC18.** follow appropriate SOP while working near any isolated and sealed off area of the mine
- PC19.** ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly
- PC20.** take relevant safety precautions during depillaring operation in UCM
- PC21.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC22.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC23.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC24.** follow laid out SOP in case of alarm signal for leakage of inflammable gases
- PC25.** follow the process of reporting any unsafe act/condition in the working area to the concerned person
- PC26.** use underground mine communication system
- PC27.** ensure positive isolation near the work place if applicable
- PC28.** use appropriate Personal Protective Equipment (PPE) as per the requirement
- PC29.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC30.** disinfect the machine/tools before and after work/task
- PC31.** maintain hygiene at the work site
- PC32.** report any symptoms of illness to the shift-incharge
- PC33.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC34.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC35.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC36.** ensure that the stowing practices produce minimum disturbance to the surface
- PC37.** ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place
- PC38.** ensure the productivity of the machine for material/fuel conservation
- PC39.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC40.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)

Qualification Pack

- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases available in the mine and their effects; and their control measures
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (Initial Medical Examination (IME) & Periodical Medical Examination (PME)) of a person employed, as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** about different types of machineries used in U/G mines
- KU11.** different types of supporting system used in U/G mines as per SSP and SSR
- KU12.** about precautions to be taken when handling heavy equipment
- KU13.** various problems/incidents likely to occur
- KU14.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS
- KU15.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU16.** common sources of pollution in the mines and ways to minimize it
- KU17.** various types of fire extinguishers
- KU18.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU19.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU20.** mining area-specific signs, and other safety and emergency signals
- KU21.** the outcome of violation of safety procedures
- KU22.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU23.** hazardous material safety, security rules and regulations
- KU24.** safety appliances and rescue equipment
- KU25.** importance and use of various communication system used in UG mines
- KU26.** importance of positive isolation at working site
- KU27.** safety and occupational health policy of organisation
- KU28.** six directional hazard identification process
- KU29.** importance of blind spot survey/training
- KU30.** basic personal and workplace hygiene
- KU31.** importance of FAB (Fresh Air Base)
- KU32.** basic provisions in Mines Creche Rules, 1966 (MCR) for any females employed in the mines
- KU33.** about basic safety regulations of Coal Mines Regulation, 2017 (CMR)
- KU34.** types of stone dust barrier and its importance
- KU35.** coal dust explosion and its preventive measures
- KU36.** classification of coal mines as per the degree of gassiness of coal seams such as first degree, second degree, and third-degree mines
- KU37.** precautions as per the gassiness of the coal mines

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- KU38.** use of flame safety lamp and its parts
- KU39.** about coal mines occupational disease such as pneumoconiosis or 'black lung' and their preventive measures
- KU40.** Standard of Ventilation as per the section 153 of the CMR 2017
- KU41.** Standard of Lighting as per the section 178 of the CMR 2017
- KU42.** the roles, duties and responsibilities of rescue team members, rescue room and rescue station and how to contact them in case of emergency
- KU43.** the correct steps for conducting any rescue work as per Mine Rescue Rule (MRR)
- KU44.** importance of taking shelter at the miner's station during blasting operation
- KU45.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU46.** importance of following infection control policies, '5-S' practices, and waste management
- KU47.** importance of water/material/energy conservation and management
- KU48.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from a machine
- GS6.** plan and organize the work order and tasks
- GS7.** use basic applications of the computer
- GS8.** use reasoning skills to identify and resolve fundamental problems
- GS9.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	43	-	14
PC1. follow preventive measures against firedamp, whitedamp, blackdamp etc.	1	1	-	-
PC2. use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)	-	1	-	-
PC3. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	1	1	-	1
PC4. comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine	1	1	-	1
PC5. ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle	-	1	-	-
PC6. ensure that every instrument, apparatus and equipment are DGMS approved before these are used	1	1	-	-
PC7. ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining	3	3	-	-
PC8. provide first aid to an injured person	1	2	-	1
PC9. follow safety precautions against spontaneous heating of the coal	1	1	-	-
PC10. operate various types of fire extinguishers to control different types of fire at worksite, if required	-	2	-	-
PC11. ensure that no person is traveling/working/staying under unsupported roof	1	-	-	-
PC12. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	-	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	-	1	-	1
PC14. use self-rescue apparatus appropriately when required	-	2	-	-
PC15. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	1
PC16. follow precautions against U/G electrical appliances	2	2	-	1
PC17. take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports	1	2	-	-
PC18. follow appropriate SOP while working near any isolated and sealed off area of the mine	1	1	-	1
PC19. ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly	1	1	-	-
PC20. take relevant safety precautions during depillaring operation in UCM	-	2	-	1
PC21. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	1	1	-	-
PC22. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	-	1	-	1
PC23. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	-	1	-	1
PC24. follow laid out SOP in case of alarm signal for leakage of inflammable gases	1	1	-	-
PC25. follow the process of reporting any unsafe act/condition in the working area to the concerned person	-	1	-	1
PC26. use underground mine communication system	-	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure positive isolation near the work place if applicable	1	1	-	-
PC28. use appropriate Personal Protective Equipment (PPE) as per the requirement	2	2	-	1
PC29. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	-	1	-	-
PC30. disinfect the machine/tools before and after work/task	1	1	-	-
PC31. maintain hygiene at the work site	1	1	-	-
PC32. report any symptoms of illness to the shift-incharge	1	1	-	-
PC33. identify six directional hazards at workplace and take decisions accordingly	-	1	-	1
<i>Follow environmental guidelines</i>	6	7	-	6
PC34. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	1	-	1
PC35. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	1	1	-	1
PC36. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-
PC37. ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place	1	1	-	1
PC38. ensure the productivity of the machine for material/fuel conservation	1	1	-	1
PC39. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	-	1	-	1
PC40. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	1	-	1



Skill Council For Mining Sector



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1704
NOS Name	Follow Health, Safety, and Environmental guidelines for underground coal mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

MIN/N1705: Follow Health, Safety and Environmental guidelines for Rare Earth (RE) Chemical plant

Description

This unit is about adhering to health, safety and environmental guidelines critical in RE chemical plants

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** comply with occupational health and safety regulations adopted by the employer
- PC2.** follow the correct safety steps in case of accident or major failure
- PC3.** operate various grades of fire extinguishers
- PC4.** follow general safety precautions while handling cables; working near electrical installations, overhead lines and while working with various electrical equipment in the plant
- PC5.** provide first-aid to an injured person
- PC6.** use appropriate PPE as per the requirement
- PC7.** identify the hazards and risks
- PC8.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC9.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC10.** disinfect the site/panel/tools before and after work/task
- PC11.** maintain hygiene at the work site
- PC12.** report any symptoms of illness to the shift-incharge
- PC13.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC14.** identify the environmental impact of operations related to RE chemical plant and take steps to reduce the impact
- PC15.** ensure the productivity of the machine for material/fuel conservation
- PC16.** follow the process for collecting, storing, and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines and safety guidelines as prescribed by regulatory authorities like Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** the outcome of violation of safety procedures
- KU2.** duties and rights of workers
- KU3.** process for reporting any unsafe act/condition in work area
- KU4.** different types of RE chemical factories and detail of the plant one is working in
- KU5.** fencing, guarding, spillage control, etc.
- KU6.** importance of first aid and hygiene
- KU7.** importance of blind spot survey/training
- KU8.** provision of wages, working hours and accident compensation as per Atomic Energy Factory Rules
- KU9.** various types of chemical processes carried out in the plant
- KU10.** RE chemical plant safety procedures
- KU11.** about installation and handling of safety devices
- KU12.** code of practice for safe handling and transport of dangerous material and heavy equipment
- KU13.** safety and occupational health policy of organisation
- KU14.** basic personal and workplace hygiene
- KU15.** six directional hazard identification process
- KU16.** Internal Safety Organisation and role of safety committee, workman's inspector and AERB etc.
- KU17.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU18.** importance of following infection control policies, '5-S' practices, and waste management as prescribed by regulatory authorities like Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.
- KU19.** importance of water/material/energy conservation and management
- KU20.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's job role
- GS2.** make decisions on the concerned area of work
- GS3.** read and explain manuals, health, and safety instructions, memos, etc.
- GS4.** plan and organize the work order and jobs
- GS5.** use reasoning skills to identify and resolve fundamental problems
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	26	44	-	17
PC1. comply with occupational health and safety regulations adopted by the employer	2	3	-	1
PC2. follow the correct safety steps in case of accident or major failure	2	4	-	1
PC3. operate various grades of fire extinguishers	2	4	-	1
PC4. follow general safety precautions while handling cables; working near electrical installations, overhead lines and while working with various electrical equipment in the plant	3	4	-	2
PC5. provide first-aid to an injured person	3	4	-	2
PC6. use appropriate PPE as per the requirement	2	5	-	2
PC7. identify the hazards and risks	3	6	-	1
PC8. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	3	5	-	2
PC9. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	1	-	1
PC10. disinfect the site/panel/tools before and after work/task	2	2	-	1
PC11. maintain hygiene at the work site	1	2	-	1
PC12. report any symptoms of illness to the shift-incharge	1	2	-	1
PC13. identify six directional hazards at workplace and take decisions accordingly	1	2	-	1
<i>Follow environmental guidelines</i>	4	6	-	3
PC14. identify the environmental impact of operations related to RE chemical plant and take steps to reduce the impact	1	2	-	1
PC15. ensure the productivity of the machine for material/fuel conservation	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. follow the process for collecting, storing, and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines and safety guidelines as prescribed by regulatory authorities like Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1705
NOS Name	Follow Health, Safety and Environmental guidelines for Rare Earth (RE) Chemical plant
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	27/08/2024
Next Review Date	26/08/2027
NSQC Clearance Date	27/08/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1706.Undertake actions to prevent the fire situation in a mine	30	50	-	20	100	15
MIN/N1707.Conduct fire rescue operation	30	50	-	20	100	25
MIN/N1708.Conduct rescue operation from surface water inundation, underground inundation, gaseous hazards, roof fall, work at height and confined places	30	50	-	20	100	25
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	110	180	-	60	350	75

Elective: 1 Opencast

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1703.Follow Health, Safety, and Environmental Guidelines for opencast mines	30	50	0	20	100	25
Total	30	50	-	20	100	25

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Elective: 2 Underground Metal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1702.Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines	30	50	0	20	100	25
Total	30	50	-	20	100	25

Elective: 3 Underground Coal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1704.Follow Health, Safety, and Environmental guidelines for underground coal mines	30	50	0	20	100	25
Total	30	50	-	20	100	25

Elective: 4 Rare Earth Plant

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1705.Follow Health, Safety and Environmental guidelines for Rare Earth (RE) Chemical plant	30	50	0	20	100	25
Total	30	50	-	20	100	25

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.